Soham Pal

Machine Learning Engineer |Data Science

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Summary

B.Tech Computer Science Engineering student with hands-on experience in **Machine Learning, Data Science, NLP, Computer Vision, Python, and SQL**. Experienced in exploratory data analysis, predictive modeling, data visualization, and deploying interactive Streamlit applications. Built ML and deep learning projects using **Scikit-learn, TensorFlow, Keras, VGG16, and BERT**.

Skills

- **Programming:** Python, Java, SQL
- **Data Analysis:** Pandas, NumPy, Excel, Power BI, Matplotlib, Seaborn, EDA, Data Cleaning
- **Machine Learning:** Scikit-learn, Regression, Classification, Clustering, Supervised Learning, Unsupervised Learning
- **Deep Learning & NLP:** TensorFlow, Keras, CNN, VGG16, BERT, VADER
- **Computer Vision:** YOLO, VGG16, PIL
- **Databases:** MySQL, PostgreSQL, DBMS
- **Tools:** Streamlit, Git,Jupyter Notebook, VS Code, REST APIs

Experience

AI Intern |Infosys Springboard *July 2026 – Sep 2026*

- Developed an NLP-based sentiment and emotion analysis system using Python, BERT and VADER to classify emotional states from text.
- Built a personalized recommendation system with Streamlit-based emotion trend visualization.
- Integrated REST APIs and PostgreSQL to manage user data, analysis results, and recommendation history within the application.

Data Science Intern |Unified Mentor Pvt. Ltd. *June 2026 – August 2026*

- Analyzed 180,519 shipment records using Python, Pandas, NumPy and EDA and quantitative metrics to evaluate delivery performance and SLA compliance.
- Identified shipping-mode and regional delay patterns using risk-ratio and delay-gap analysis, revealing 54.83% late-delivery risk and key high-risk logistics segments.
- Developed and deployed a Streamlit dashboard using Pandas, Matplotlib, Seaborn, and Scikit-learn to visualize logistics KPIs.

Projects

Brain Tumor Detection Using Deep Learning *View Project* *March 2026*

- Developed a Deep Learning-based MRI image classification system using TensorFlow, Keras, and VGG16 to classify tumor and non-tumor cases.
- Achieved 93% classification accuracy through image preprocessing, feature extraction, model training, and evaluation.

Customer Behavior Analyst *View Project* *January 2026*

- Analyzed customer shopping behavior using Python, Pandas, MySQL, and Power BI to identify purchasing trends and customer segments.
- Developed interactive Power BI dashboards to visualize customer KPIs and support data-driven decisions.

US Flight Delay Prediction *View Project* *November 2025*

- Analyzed U.S. airline data using Python, Pandas, and EDA to identify factors associated with flight delays, cancellations, and diversions.
- Built a Scikit-learn classification model to predict flight delays from airline and operational features.

Education

B.Tech in Computer Science and Engineering *Aug 2023 – Jun 2027*
Adamas University, Kolkata **CGPA: 6.5/10**

Higher Secondary (PCMB) *Mar 2023*
Lego R.B.R.K. Vidyapith 73%

Secondary *Feb 2021*
Lego R.B.R.K. Vidyapith 82%

Certifications

- **Enterprise Design Thinking Practitioner** – IBM Certificate
- **Advance Python** – simplilearn Certificate
- **Data Analyst Course** – Physics Wallah Skills Certificate

Hackathons

Smart India Hackathon (SIH) *Sep 2025*

- Developed an ML-based crop recommendation system using soil and environmental parameters to identify suitable crops.